



Generative inverse design of bioinspired disordered metamaterials for strength-toughness synergy

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ARTICLE INFO

Keywords:

Mechanical metamaterials
Machine learning
Inverse design
Bioinspired design
Nonlinear tensile responses
Strength-toughness synergy

ABSTRACT

Natural irregularity offers an attractive route to lightweight, multifunctional, and damage tolerant mechanical metamaterials. However, tailoring nonlinear tensile responses in such systems remains challenging because of the complex relationships between structure and properties and the intrinsic trade-off between specific energy absorption and specific strength. Here, a spiderweb inspired inverse design framework for disordered metamaterials with programmable tensile behavior is reported. A virtual growth algorithm constructs disordered architectures from global prior frequency fields mapped from spiderweb patterns, while a residual SwiGLU deep neural network predicts their complex nonlinear mechanical responses. Pearson correlation and SHAP analyses further uncover the structure-property relationships and physical mechanisms underlying performance. To achieve inverse design, a surrogate assisted conditional generative adversarial network is integrated with Monte Carlo dropout, genetic algorithms and Bayesian optimization, enabling diverse and physically meaningful solutions to the one-to-many mapping problem. Following FEA validation, several generated disordered architectures remain beyond the pre-optimization Pareto front. The fabricability of three representative scaled architectures is further confirmed by tensile tests, and qualitative agreement is observed between the simulated and experimental crack paths. These results provide a systematic strategy for tailoring nonlinear tensile responses and developing disordered metamaterials with balanced SEA and SS.

1. Introduction

Mechanical metamaterials derive their exceptional properties primarily from their architectural designs, not from modifications to their chemical composition [1–3]. By encoding function into geometry, they have enabled unusual mechanical behaviors, including programmable stiffness [4,5], enhanced energy absorption [6–8], and negative Poisson's ratio [9–11], with applications ranging from aerospace structures [12] to biomedical scaffolds [13] and soft robotics [14]. Most established metamaterial designs, however, rely on periodic architectures [15]. Although periodicity simplifies fabrication and analysis, it also imposes uniform load transfer paths that can amplify stress concentrations under tensile loading, accelerating crack propagation and promoting brittle failure [16,17].

Natural materials suggest a different design principle. Across diverse biological systems, irregularity is not a defect but a source of mechanical

resilience. Inspired by this principle, honeycomb and spiderweb based architectures have been explored for deformation control, energy absorption, and vibration isolation, providing a promising bioinspired strategy for metamaterial design [18–20]. Disordered porous networks in marine sponges [21], vascular bundles in bamboo [22], and hierarchical venation networks in plant leaves [23] all exhibit disordered heterogeneous features at the microscopic scale. Such disordered topologies not only redistribute loads through multiple paths to avoid local stress peaks but also promote crack deflection and branching, thereby dissipating more fracture energy [24,25]. Such mechanisms enhance damage tolerance and energy dissipation while preserving structural efficiency. Inspired by these natural structures, recent studies have begun to introduce controlled disorder into metamaterial design as well as synthesize disordered topologies through computational algorithms [26,27]. Specifically, a probabilistic virtual growth program enables the generation of stochastic microstructures governed by local

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<https://doi.org/10.1016/j.ijmecsci.2026.111961>

Received 29 April 2026; Received in revised form 27 July 2026; Accepted 29 July 2026

Available online 30 July 2026

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adjacency rules [28]. Based on this generative paradigm, subsequent studies have broadened its algorithmic framework and application scope. For instance, growth rules have been integrated with unstructured graphs and topology optimization to explore larger design spaces [29,30]. Furthermore, this paradigm has facilitated the modulation of stress distributions and the decoupled design of hybrid metamaterials via deep learning [4,31]. Recent studies have shifted from structural generation toward nonlinear inverse design. Growth based and deep learning frameworks have been developed for disordered metamaterials with target force deflection or programmable responses [32,33], whereas autoregressive and generative models have addressed target nonlinear behavior and customized stress strain curves [34,35].

Despite these advances, the inverse design of disordered metamaterials with programmable nonlinear tensile responses remains insufficiently studied. More broadly, previous studies have examined geometric symmetry in topologically interlocked systems [36], inverse design of growth-inspired irregular architectures [37], and machine-learning-assisted tuning of re-entrant honeycombs [38]. These studies provide useful methodological context for structural design but focus mainly on geometric generation or selected response quantities. Tensile behavior is more difficult to control because it emerges from the coupled evolution of large deformation, local stress concentration, damage initiation and progressive fracture [39,40]. The tensile stress strain curve therefore encodes a far richer mechanical process than scalar descriptors alone [41,42]. Resolving these processes for large numbers of irregular architectures remains computationally expensive. Adaptive coarse grained formulations have reduced the cost for notched 2D and 3D spinodal metamaterials [43,44], while experimental studies of 3D printed twisting tubular metamaterials illustrate the practical burden of fabricating and testing broad structural families [45].

The computational and experimental costs limit the exploration of disordered metamaterials, particularly when multiple mechanical objectives need to be optimized jointly. For lightweight metamaterials under tensile loading, specific energy absorption (SEA) and specific strength (SS) are both important, but they are often regarded as competing metrics [46,47]. Discovering architectures that improve both simultaneously requires efficient exploration of a vast and highly non-convex space of disordered topologies [48]. Machine learning has expanded the exploration of deformation-dependent Poisson responses and kirigami mechanics [49,50], stress evolution, crack paths, and fatigue life [51,52], as well as lattice anisotropy and mechanical performance [53,54]. Related models have been developed for programmable diamond plate lattices and complex metamaterials represented by multimodal inputs [55,56]. However, many data-driven studies remain focused on periodic or compression-dominated systems. Representative examples include the prediction of compressive stress-strain curves and the customization of lattice energy absorption [57,58], as well as generative optimization of structural lattices [59]. Equivalent frameworks for disordered systems under tensile loading remain scarce. Furthermore, customizing these metamaterials to achieve targeted mechanical responses requires inverse design [60–63]. This process faces an inherent one-to-many mapping issue, where distinct micro-topologies can yield highly similar nonlinear stress-strain curves [64,65]. As a result, traditional deterministic deep learning models often collapse onto invalid or physically disconnected structural layouts, limiting their practical utility.

This challenge motivates hybrid inverse design workflows that integrate candidate generation, efficient response evaluation, and refinement based on optimization. Similar strategies driven by target responses have been explored in acoustic and vibration metamaterials. In these studies, an inverse network based on an autoencoder is used to generate candidate structures for broadband sound absorption, and rapid band gap evaluation is coupled with optimization techniques for vibration absorbing elastic metamaterials [66,67]. Although the underlying physical responses differ, these studies illustrate the use of hybrid workflows driven by targets in metamaterial design. For the

present quasi static tensile design problem, a spiderweb inspired framework is proposed for the inverse design of disordered metamaterials with tailored nonlinear tensile behavior. Natural spider webs illustrate how spatial architecture and nonlinear silk behavior jointly contribute to mechanical robustness. Cranford et al. showed that this robustness arises from the coupling between web geometry and mechanically distinct silk threads, including relatively stiff radial dragline threads and more compliant spiral threads [68]. In the present study, the spiderweb pattern serves as a geometric prior for spatial frequency field construction, not as a direct model of spider-silk mechanics. On this basis, the proposed framework is implemented through the following steps. First, a virtual growth algorithm (VGA), guided by global prior frequency fields mapped from spiderweb patterns, is utilized to automatically assemble and generate a large library of disordered metamaterials, thereby constructing an expansive and parameterized design space. Subsequently, a residual SwiGLU deep neural network (RS-DNN) is developed as a forward surrogate model to predict nonlinear tensile stress-strain curves with high efficiency. Third, interpretable learning tools, including Pearson correlation and SHAP analysis, are utilized to quantitatively reveal the underlying influence of geometric and frequency parameters on macroscopic performance metrics, specifically SEA and SS. Fourth, to resolve the one-to-many mapping issue, a surrogate-assisted conditional generative adversarial network (SA-cGAN) integrated with Monte Carlo (MC) dropout, genetic algorithm (GA), and Bayesian optimization (BO) is proposed to generate and refine diverse candidate structures for prescribed responses.

Three advances are provided by this framework. A scalable route is established for modeling and inverse designing disordered metamaterials under tensile loading, a regime that remains underexplored relative to compression. Generative design is connected with interpretable analysis to reveal the structural features that control the strength toughness balance. The discovery of unconventional architectures that extend beyond the pre-optimization Pareto front of SEA and SS is also enabled. Together, these results provide a general strategy for designing lightweight disordered metamaterials whose tensile responses can be programmed beyond conventional trade-offs.

The remainder of this paper is organized as follows. Section 2 describes the spiderweb pattern informed architecture generation process and the FEA based mechanical database. Section 3 presents the RS-DNN forward surrogate and the SA-cGAN inverse design workflow with uncertainty evaluation and iterative refinement. Section 4 reports the prediction and inverse design results, interpretable analyses, FEA validation, periodic reference comparison, and experimental feasibility assessment. Section 5 summarizes the main findings, limitations, and directions for future work.

2. Design and mechanical characterization of spiderweb-inspired disordered metamaterials

This section describes how spiderweb pattern informed disordered architectures are generated and converted into a mechanical response database. Section 2.1 introduces the spatial prior, frequency constraints, and connectivity controlled VGA, while Section 2.2 presents the FEA model, PA12 nylon material setting, damage formulation, and extraction of nonlinear tensile stress-strain curves.

2.1. Spiderweb-inspired generation of disordered architectures

To achieve tunable nonlinear mechanical responses in disordered metamaterials, a virtual growth algorithm (VGA) design framework is proposed in this study, in which spiderweb inspired frequency fields are used to guide structural generation. This framework uses the macroscopic morphology of a spiderweb as a spatial prior for microstructural assembly. The generated building blocks are not treated as analogues of radial or spiral silk threads, and the mapping is not designed to represent the material differentiation or nonlinear behavior of natural spider silk.

Therefore, the resulting architectures are spiderweb-pattern-informed designs and do not mechanically replicate biological spiderwebs. As illustrated in Fig. 1(a), the algorithm initially constructs a continuous domain pattern of a spiderweb utilizing prescribed geometric parameters such as layers (N_l), angle (θ), spiral width (W_s) and radial width (W_r), which is subsequently discretized and mapped onto a 25×25 spatial grid matrix. Given that previous studies have demonstrated that the nonlinear mechanical properties of metamaterials tend to stabilize when the grid size reaches 20×20 , the 25×25 grid size selected here not only ensures the stability of the mechanical responses but also provides a diverse design space for the micro-topological configurations [28].

After spatial discretization, the grid is precisely divided into two topological subdomains based on the pixel grayscale threshold: the web region (denoted by subscript w, represented by black pixels) and the background region (denoted by subscript b, represented by white pixels). Subsequently, for these two independent subdomains, the occurring frequencies of the four fundamental building blocks, namely Cross (C), Line (L), X, and T, are parameterized independently as f_i^j (where $i \in \{w, b\}$ and $j \in \{C, L, X, T\}$). Through the explicit assignment of these block frequencies across different regions, a global prior frequency field is ultimately constructed by the algorithm to guide the micro-assembly, as shown in Fig. 1(b). Table 1 summarizes the independent variables governing the design space of the bioinspired metamaterials. These include four geometric parameters of the spiderweb (N_l , θ , W_s , W_r) and one core frequency parameter (the distribution frequency of the Cross unit within the web region, denoted as f_w^C). To ensure the normalization of generative frequencies across different domains, the remaining seven frequency parameters are explicitly derived from f_w^C . The specific constraint relationships are defined as follows: $f_w^L = f_w^T = f_b^C = f_b^X = 0.5 - f_w^C$ and $f_w^X = f_b^L = f_b^T = f_w^C$. These relationships reduce the number of independent frequency variables while maintaining a controlled frequency contrast between the web and background regions. Because all frequency variables are derived from a

Table 1

Design space of morphological and frequency parameters for the spiderweb-inspired virtual growth framework.

Parameter	Symbol	Value	Number
Number of layers	N_l	8, 9, 10, 11, 12	5
Angle	θ	30, 36, 40, 45	4
Spiral width	W_s	4, 5, 6	3
Radial width	W_r	4, 5, 6	3
Cross frequency	f_w^C	0.27, 0.29, ..., 0.47, 0.49, 0.50	13

single core parameter, the available block frequency combinations are coupled and are not varied independently. Therefore, the design space does not cover independently assigned block frequencies, frequency settings without complementary relations between the web and background regions, or local frequency variations within each region. Together with the four geometric variables, this parameterization yields 2,340 unique architectures within the design family informed by spiderweb patterns. Thus, the database is limited to this prescribed design family and does not cover unconstrained frequency fields.

After the target frequency distribution matrix is obtained, the unit cells within the 2D grid are progressively filled and assembled by the VGA. Fig. 2(a) presents the four fundamental building blocks and their rotated variations required to construct the metamaterial architectures. Each block has an overall dimension of $1.2 \text{ mm} \times 1.2 \text{ mm}$ and a strut width of 0.33 mm . The introduction of these variations significantly enriches the block library and broadens the design space, thereby endowing the generated structures with more complex and diverse nonlinear mechanical responses.

When determining the specific building block for a given location, the algorithm is strictly governed by physical constraints and frequency guidance. During the structural growth phase, the system prioritizes the grid with the minimum entropy, indicating that this grid possesses the fewest available candidate blocks [28,30]. Subsequently, based on the pool of admissible variations, the system assigns a candidate block that complies with the adjacency rules. This physical constraint mechanism

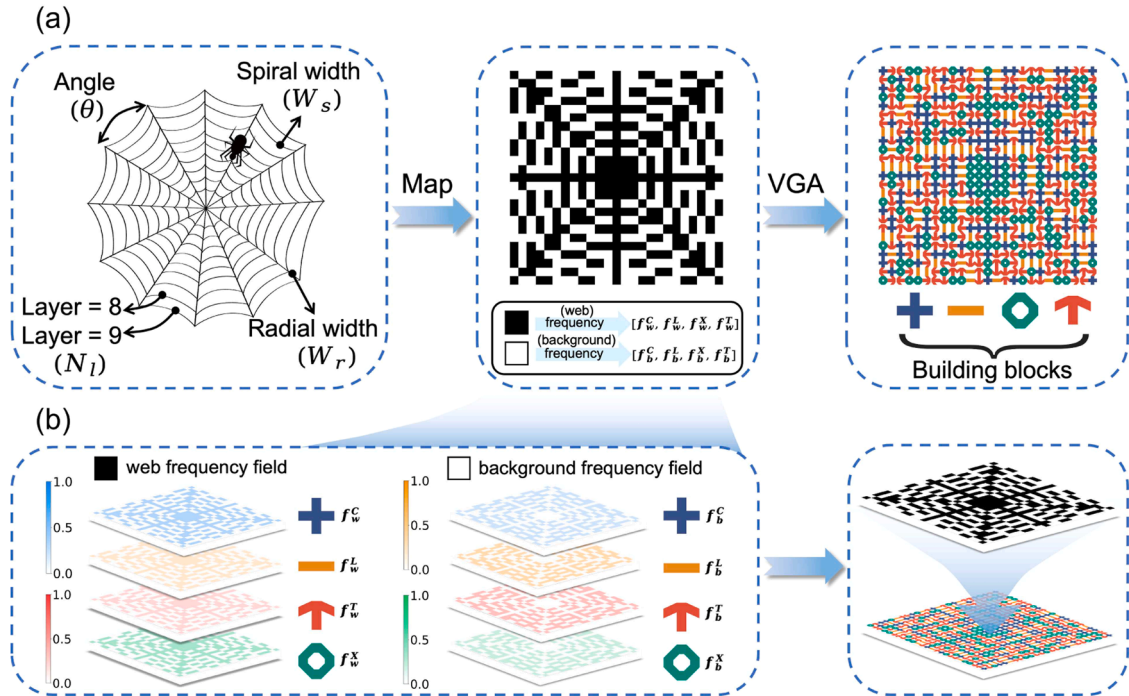


Fig. 1. Spiderweb-pattern-informed morphological transformation and prior frequency-field generation. (a) Parametric generation and spatial discretization, where a continuous-domain spiderweb pattern is constructed using twelve design parameters and subsequently mapped onto a 25×25 spatial grid. (b) Frequency mapping and blueprint construction, where the discrete grid is divided into web and background regions to explicitly assign block frequencies, ultimately constructing a global prior frequency field to guide the microstructural assembly.

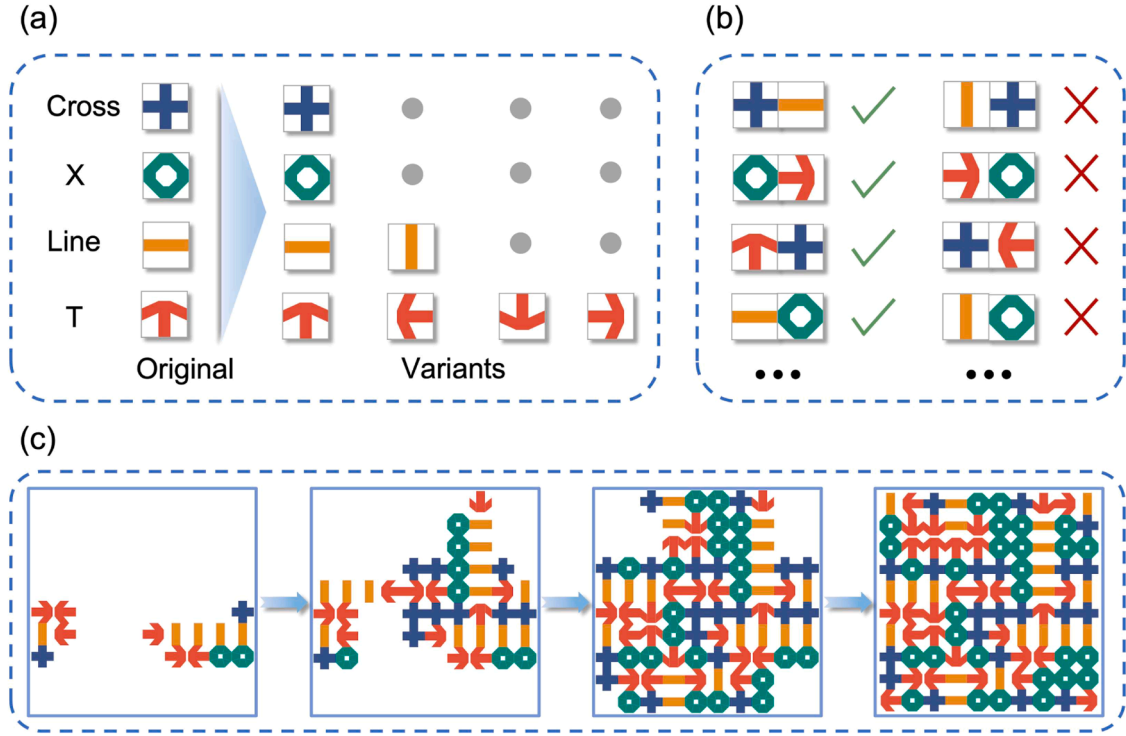


Fig. 2. Virtual growth algorithm and micro-assembly process. (a) Fundamental building blocks, presenting the four basic units and their rotated variations that significantly enrich the block library and broaden the design space. (b) Adjacency rules and physical constraints, illustrating the allowed and prohibited boundary connections designed to strictly forbid the generation of disconnected sub-regions. (c) Iterative virtual growth process, demonstrating the frequency-guided block allocation process to guarantee the global connectivity of the generated architecture.

ensures proper connectivity between adjacent building blocks, effectively avoiding the generation of disconnected sub-regions (allowed and prohibited adjacency cases are detailed in Fig. 2(b)). Simultaneously, the selection frequencies of these candidate blocks are evaluated by the algorithm in real time by incorporating the previously established prior frequency distribution, prioritizing the building block that best aligns with the global frequency mapping target.

Ultimately, relying on these rules, the appropriate building block type and its orientation are precisely allocated to each grid by the system. Through rigorous local physical boundary screening and frequency-guided iterative optimization, a metamaterial architecture featuring fully connected micro-geometry and a macroscopic density exhibiting spiderweb morphological characteristics is successfully generated by the algorithm. Fig. 2(c) illustrates the complete virtual growth process. To prevent growth interruption, if the placement frequency of any grid converges to zero, a backtracking mechanism is triggered by the algorithm to retract the iteration step, reverting to the previous grid to reallocate options, which ultimately ensures the global connectivity of the architecture.

2.2. Computational characterization of nonlinear tensile responses

Finite element simulations were performed to obtain the tensile stress-strain responses of the disordered metamaterial structures. A Python-based scripting approach is utilized to automate the batch simulation workflow. As illustrated in Fig. 3(a), the top and bottom boundaries of the structure are tied to their respective reference points (RPs) using kinematic coupling. The top RP is prescribed a displacement of 4 mm applied via a smooth step amplitude curve, with all degrees of freedom constrained except for the loading direction; meanwhile, the bottom RP is fully constrained in all degrees of freedom. The structure is discretized using CPE4R elements (four-node, bilinear plane-strain

elements with reduced integration and enhanced hourglass control) and CPE3 elements (three-node plane-strain triangles), which are highly effective in capturing in-plane deformation and fracture behavior. After the mesh convergence analysis provided in Fig. 3(b), an element size of 0.07 mm is adopted, which optimally balances the trade-off between computational efficiency and numerical accuracy. The base material is defined as an elastoplastic solid representative of commonly used PA12 nylon, with a mass density ρ of 1,356 kg/m³. The elastic region is characterized by a Young's modulus E of 1,993 MPa and a Poisson's ratio ν of 0.4. In the plastic region, an initial yield stress σ_y of 46.0 MPa is specified. PA12 nylon was selected as a representative additively manufacturable thermoplastic because it provides moderate stiffness and strength, finite ductility, nonlinear elastic-plastic deformation, and progressive damage behavior. These characteristics allow the generated architectures to exhibit load redistribution, local bending, node rotation, damage initiation, and fracture under tensile loading. In addition, recent experimental work on disordered mechanical metamaterials has used printable nylon materials to study regions prone to fracture and the enhancement of strength and toughness, supporting the use of PA12 nylon as a practical polymer testbed for metamaterial design involving fracture [69]. Crack propagation is explicitly modeled through a ductile damage framework, initiated at a fracture strain of 0.1 under a stress triaxiality of 0.333. Subsequently, material failure is governed by an energy-based damage evolution with a fracture energy of 0.8 kJ/m², and the element deletion capability is activated at a maximum degradation factor of 0.8 to physically represent the fracturing process. Finally, general self-contact is incorporated to account for interactions between the structural walls during large deformations. The contact behavior is modeled using a penalty-based friction formulation in the tangential direction, with a friction coefficient of 0.8. Fig. 3(c) illustrates the stress-strain curves of all disordered metamaterial structures within the training set, demonstrating the accessible design space. Fig. 3(d)

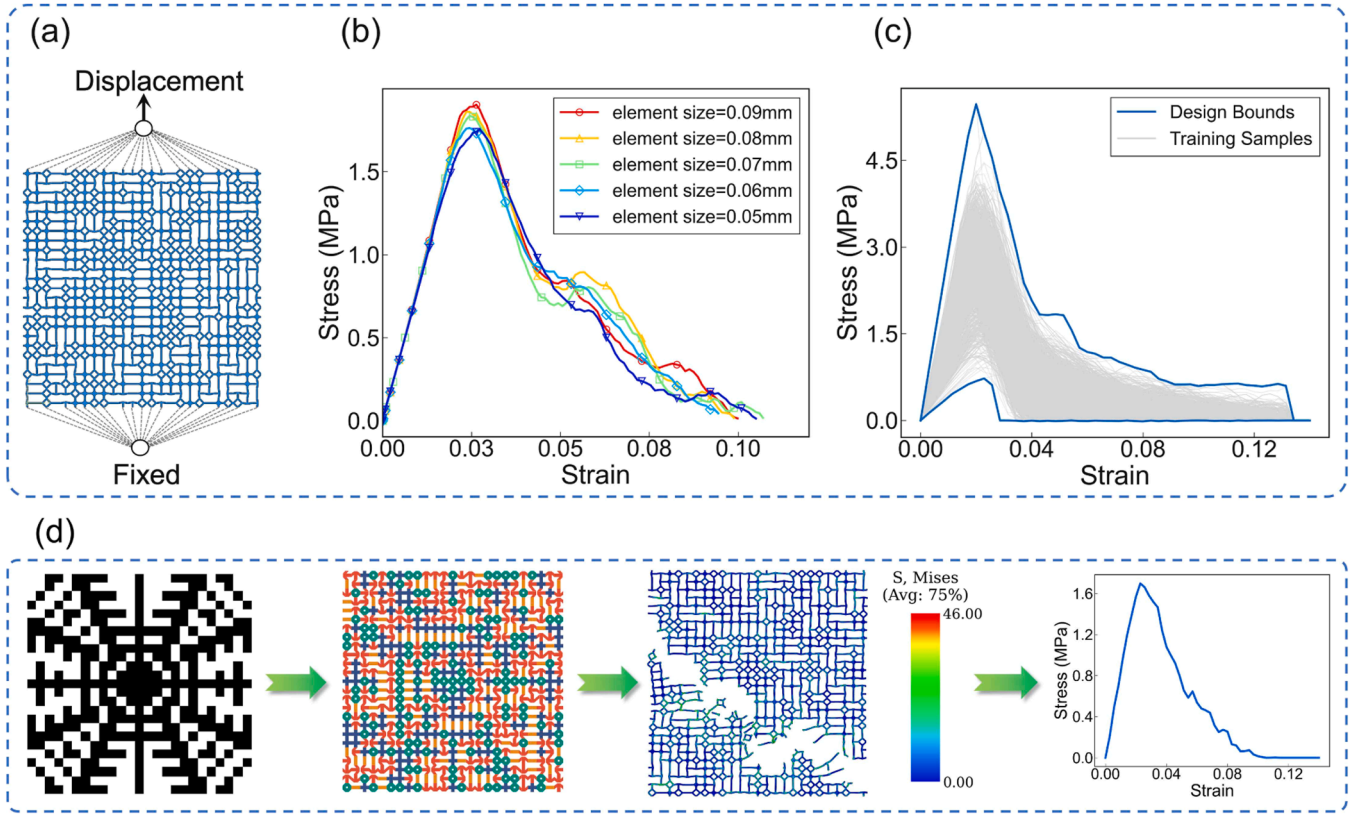


Fig. 3. Finite element setup and computational pipeline. (a) Schematic of the finite element setup, where the bottom of the structure is fully fixed and a displacement is applied to the top. (b) Mesh convergence analysis, demonstrating that the stress-strain curves are consistent across different mesh sizes. (c) Mechanical design space, where gray curves represent the stress-strain curves of the generated training samples, and blue curves outline the upper and lower performance bounds of disordered structures. (d) Mechanical properties computational pipeline, the tensile fracture mechanism is evaluated by finite element analysis (FEA) and the corresponding nonlinear stress-strain curve is extracted.

demonstrates the computational pipeline from micro-topological reconstruction to finite element analysis (FEA), ultimately extracting the nonlinear stress-strain curve. Together, the high-fidelity mechanical dataset required for subsequent surrogate modelling and inverse design is provided by this simulation framework.

3. Deep learning framework for forward prediction and uncertainty-aware inverse design

This section presents the learning framework used for forward prediction and inverse design. Section 3.1 describes the RS-DNN architecture, training objective, and evaluation metrics. Section 3.2 introduces SA-cGAN and the surrogate evaluated mechanical loss, and Section 3.3 explains MC dropout, direct candidate selection, and refinement using GA and BO.

3.1. Architecture and training strategy of the forward prediction network

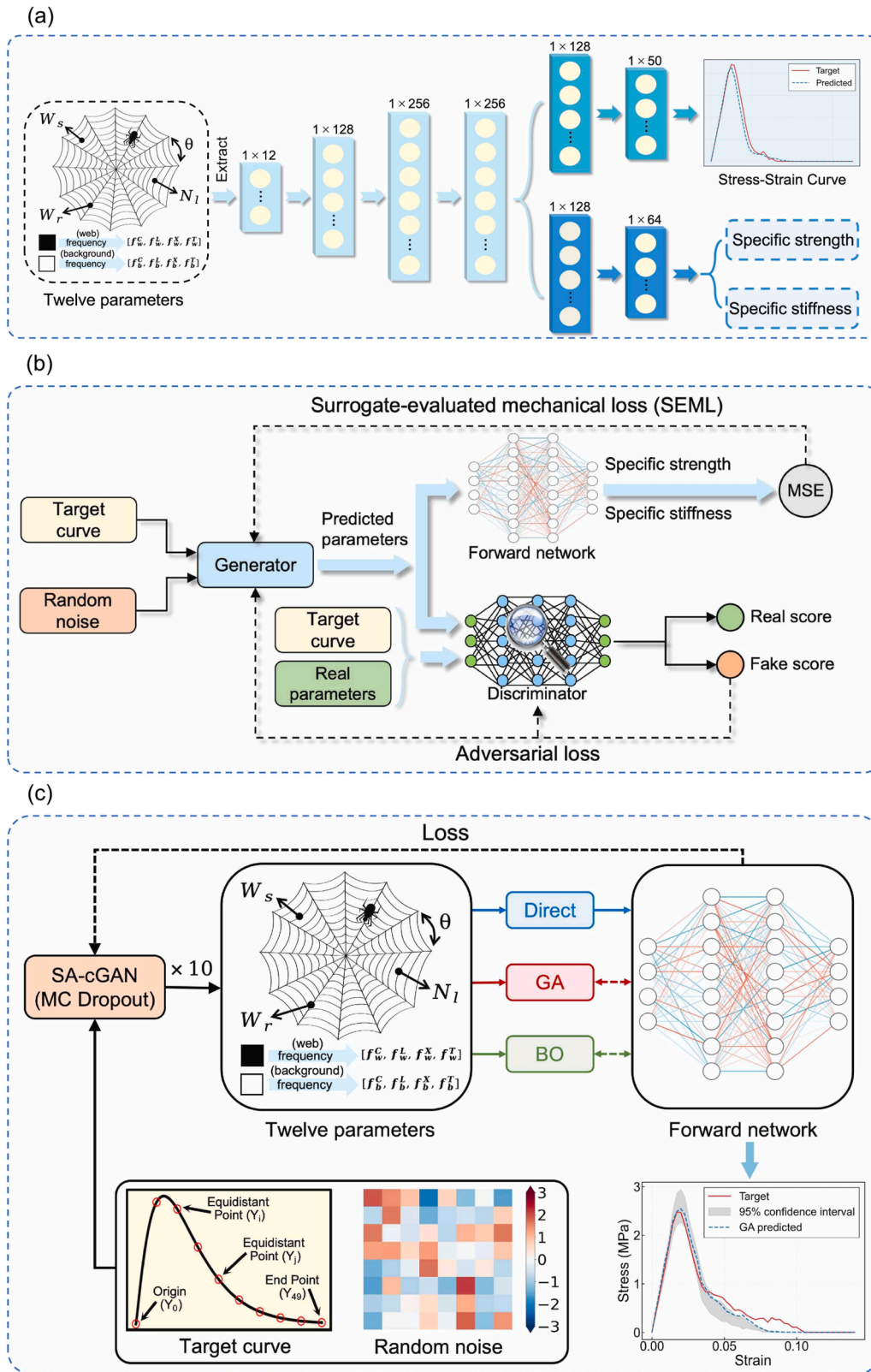
To predict the highly non-linear mechanical responses of spiderweb-inspired metamaterials, a multi-task deep neural network termed the residual SwiGLU deep neural network (RS-DNN) is developed. As shown in Fig. 4(a), the network takes 12 design parameters as input and outputs a 50-dimensional stress-strain curve alongside specific strength (SS) and specific stiffness (ST). To standardize the output representation across samples with different fracture strains, all force-displacement curves were first converted into stress-strain curves. Fifty stress values were then sampled at uniformly spaced strain intervals within the effective truncated strain range using cubic interpolation. This preprocessing yields a consistent representation of the nonlinear tensile response suitable for supervised learning.

The RS-DNN network architecture features shared hidden layers and two task-specific prediction heads, comprising multiple residual SwiGLU blocks and dropout layers to improve training stability and mitigate overfitting. Detailed hyperparameters are summarized in Table 2. To train and evaluate the model, the full dataset is first split into training and testing subsets using a 9:1 ratio. Five-fold cross-validation is then conducted on the training set, and the selected model is finally evaluated on the unseen testing set to assess its generalization capability.

To jointly learn the continuous stress-strain curve and the scalar mechanical quantities, an adaptive multi-task objective based on homoscedastic uncertainty is employed. First, for curve prediction, since the stress generally drops to zero during the large strain stage (post-fracture), a masked mean squared error loss (L_{curve}) is designed to guide the network to focus more on the effective load-bearing phase of the material. Second, for the prediction of SS and ST, a standard mean squared error loss (L_{ss} and L_{st}) is utilized. Lastly, a relative error loss (L_{rel}) is integrated to enforce physical fidelity in energy absorption (EA). The resulting composite loss therefore balances pointwise curve accuracy, scalar prediction fidelity and physically meaningful energy characterization within a unified optimization framework. The definitions of those loss terms are given as follows:

$$L_{curve} = \frac{\sum_{i=1}^n \sum_{j=1}^m M_{ij} \cdot (\hat{y}_{curve}^{(ij)} - y_{curve}^{(ij)})^2}{\sum_{i=1}^n \sum_{j=1}^m M_{ij} + \epsilon}, \quad (1)$$

$$L_{ss} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_{ss}^{(i)} - y_{ss}^{(i)})^2, \quad (2)$$



(caption on next page)

Fig. 4. Deep learning framework. (a) Architecture and training strategy of the multi-task residual SwiGLU deep neural network (RS-DNN). The network includes shared hidden layers with residual SwiGLU blocks and two task-specific prediction heads. The RS-DNN inputs 12 design parameters and outputs a 50-dimensional stress-strain curve alongside specific strength (SS) and specific stiffness (ST). (b) Architecture of the surrogate-assisted cGAN (SA-cGAN) for inverse design. The framework includes a generator, a discriminator, and a pre-trained forward surrogate model (RS-DNN). The generator inputs the target stress-strain curve alongside random noise and outputs geometric and frequency parameters. The discriminator inputs generated and authentic parameters to output the adversarial loss. Simultaneously, the RS-DNN inputs the predicted parameters and outputs SS and ST to compute the surrogate-evaluated mechanical loss (SEML), effectively enhancing physical consistency. (c) Surrogate-guided search strategy framework. The framework includes a generative inverse model (SA-cGAN), a forward surrogate model (RS-DNN), and three search pathways. The SA-cGAN inputs the target mechanical response and outputs candidate parameter sets. The RS-DNN inputs these candidate parameters and outputs predicted stress-strain curves. The search pathways input the evaluated candidates and output the ultimate geometric and frequency parameters through direct selection, genetic algorithm (GA) and Bayesian Optimization (BO).

Table 2
Neural network architecture and hyperparameters of RS-DNN.

Hyperparameters	Shared hidden layers	Curve head	Scalar head
Input dimensions	12	256	256
Hidden dimensions	128, 256, 256	128	128, 64
Output dimension	256	50	2
Activation function	SwiGLU	SwiGLU, Sigmoid	SwiGLU, Sigmoid
Optimization algorithm	AdamW	AdamW	AdamW
Batch size	64	64	64
Learning rate	10^{-5}	10^{-5}	10^{-5}
Training epoch	1000	1000	1000
Dropout	0.10	0.10	0.10

$$L_{st} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_{st}^{(i)} - y_{st}^{(i)})^2, \quad (3)$$

$$L_{rel} = \frac{1}{n} \sum_{i=1}^n \frac{\int |\hat{y}_{curve}^{(i)} - y_{curve}^{(i)}| dx}{\int y_{curve}^{(i)} dx + \epsilon}, \quad (4)$$

$$L_{total} = L_{curve} + \sum_{k \in \{ss, st, rel\}} \left(\frac{1}{2\sigma_k^2} L_k + \log \sigma_k \right), \quad (5)$$

where n is the number of samples in a mini-batch, m is the number of extracted data points on the stress-strain curve, and M_{ij} is the specific mask weight assigned to the j -th point of the i -th sample to prioritize the effective load-bearing phase. $\hat{y}_{curve}^{(ij)}$ and $y_{curve}^{(ij)}$ are the predicted and ground-truth stress values at the j -th point, respectively. $\hat{y}_{curve}^{(i)}$ and $y_{curve}^{(i)}$ denote the continuous predicted and ground-truth stress-strain curves for the i -th sample. ϵ is an infinitesimally small constant to prevent zero division. For the physical scalars, $\hat{y}_{ss}^{(i)}$ and $y_{ss}^{(i)}$ represent the predicted values and ground-truth of SS, $\hat{y}_{st}^{(i)}$ and $y_{st}^{(i)}$ represent the predicted values and ground-truth of ST. σ_k denotes the learnable homoscedastic uncertainty parameter corresponding to each task for dynamic gradient balancing.

The predictive performance of the RS-DNN is evaluated with the metrics of coefficient of determination (R^2), mean squared error (MSE), and normalized RMSE ($NRMSE$). These metrics provide complementary measures of goodness of fit, absolute prediction error, and scale-normalized accuracy. The definitions of these metrics are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (6)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (7)$$

$$NRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}}{y_{\max} - y_{\min}}, \quad (8)$$

where y_i is the ground-truth value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the true values, and n is the number of samples. $y_{\max}$ and $y_{\min}$ correspond to the maximum and minimum values of the ground-truth dataset for a specific physical quantity.

3.2. Surrogate-assisted inverse design of disordered metamaterials

To establish an inverse mapping from the desired nonlinear mechanical responses to disordered metamaterial architectures, a surrogate-assisted conditional generative adversarial network (SA-cGAN) architecture is constructed, as illustrated in Fig. 4(b). The generator network takes as input a target nonlinear stress-strain curve with a random noise vector drawn from a standard normal distribution and maps them into the corresponding geometric and frequency parameters that define the architecture. At the output layer of the generator, Softmax activation functions are applied to the frequency prediction heads corresponding to the two independent topological subdomains (i.e., the web region and the background region). This mechanism explicitly ensures the normalization of generative frequencies across different domains, thereby preserving the intrinsic physical validity of the generated micro-topological configurations from the source.

The generated geometric and frequency parameters undergo rigorous evaluation through a dual-feedback mechanism. On one hand, the discriminator compares the generated parameters with the authentic dataset to compute the adversarial loss, which guides the generator to produce parameters that conform to the prior distribution of real physical designs. On the other hand, to enhance the physical consistency of the generated designs, the framework integrates a pre-trained and weight-frozen forward surrogate model (RS-DNN). The surrogate model receives the predicted parameters and rapidly predicts the scalar mechanical quantities specific strength (SS) and specific stiffness (ST). The mean squared error (MSE) between these surrogate-predicted properties and their corresponding ground-truth targets is defined as the surrogate-evaluated mechanical loss (SEML). By incorporating SEML as a physics-based regularization term into the adversarial training, the generator is guided to produce structures whose surrogate-evaluated mechanical properties closely match the target values. This SA-cGAN framework therefore couples distribution matching with property matching, enabling the generation of realistic disordered microgeometries that approximate the target nonlinear mechanical behavior. The individual and combined effects of the SEML and MC dropout on target matching and generative diversity are further examined through a controlled ablation study in Supplementary Note S3. To formulate the above dual-feedback optimization process, the core loss functions utilized in the SA-cGAN are defined as follows:

$$L_D = \frac{1}{2} [BCE(D(x, c), 0.9) + BCE(D(G(z, c), c), 0.0)], \quad (9)$$

$$L_{adv} = BCE(D(G(z, c), c), 1.0), \quad (10)$$

$$L_{SEML} = \frac{1}{N} \sum_{i=1}^N \left(F_{RS-DNN}(G(z^{(i)}, c^{(i)})) - S_{target}^{(i)} \right)^2, \quad (11)$$

$$L_G = L_{adv} + \left(\frac{1}{2\sigma_{phys}^2} L_{SEML} + \log \sigma_{phys} \right), \quad (12)$$

where N is the number of samples in a mini-batch; $\mathbf{x}$ is the real parameter vector; $\mathbf{c}$ is the target stress-strain curve serving as the condition; and $\mathbf{z}$ is the random noise vector. $G(\cdot)$ and $D(\cdot)$ denote the mapping functions of the generator and discriminator, respectively. BCE represents the Binary Cross-Entropy loss, which incorporates a label smoothing strategy (real targets set to 0.9 and generated targets to 0.0) to enhance the stability of adversarial training. L_{adv} is the adversarial penalty term driving the generator to produce more plausible parameters. In the surrogate feedback module, F_{RS-DNN} denotes the frozen forward surrogate model, and $S_{target}^{(i)}$ represents the target physical scalars (SS and ST) for the i -th sample. Finally, σ_{phys} is the learnable homoscedastic uncertainty parameter used in the composite loss L_G to dynamically and adaptively balance the gradients between adversarial distribution fitting and physical precision alignment. The training stability and hyperparameter sensitivity of SA-cGAN were further evaluated in Supplementary Note S4, where convergence behavior, adversarial loss, and the effects of noise-vector dimension and label-smoothing value were examined.

3.3. Uncertainty-aware inverse design and iterative refinement pathways

In the inverse design of disordered metamaterials, multiple distinct micro-topological configurations can yield highly similar nonlinear mechanical properties, a challenge widely known as the one-to-many mapping issue [70,71]. To address this issue while enabling efficient exploration of the expansive design space, a surrogate-guided search strategy that couples the SA-cGAN with the RS-DNN is proposed, as illustrated in Fig. 4(c). Given a target mechanical response, the SA-cGAN acts as a generative inverse model. To evaluate the network-inherent uncertainty, the Monte Carlo (MC) dropout mechanism is activated during the inverse design stage. This approach quantifies the prediction fluctuations and confidence levels of the neural network arising from limited training data and parameter approximations. Under this uncertainty-aware setting, the SA-cGAN produces ten distinct candidate parameter sets, including both geometric and frequency parameters. This allows the model to map the desired nonlinear mechanical properties not into a single deterministic solution, but into a dependable prior distribution space of candidate solutions.

However, data-driven generative models occasionally yield parameters that deviate from the intrinsic physical validity of the architectures. To efficiently evaluate the physical consistency of the generated parameters while bypassing the expensive finite element analysis (FEA), the pre-trained RS-DNN is employed as a forward surrogate model. Within this collaborative architecture, three distinct methodological pathways are established to derive the ultimate geometric and frequency parameters. In the baseline direct selection pathway, the RS-DNN explicitly predicts the stress-strain curves for the ten initial candidates. A filter-and-select mechanism then isolates the single design that minimizes the normalized root mean square error (NRMSE) against the target curve, requiring no further iterative optimization.

To further reduce prediction errors and improve the physical consistency of the generated designs, the framework introduces two distinct refinement pathways. In the SA-cGAN and genetic algorithm (GA) pathway, the ten generated candidates constitute the initial population. Guided by the RS-DNN-evaluated fitness scores, the GA performs iterative selection, crossover, and mutation, systematically searching the localized design space to refine the geometric and frequency parameters. Furthermore, the SA-cGAN and Bayesian Optimization (BO) pathway utilizes these candidates as high-quality initial observation points to construct a Gaussian Process (GP) surrogate. By leveraging an expected improvement acquisition function, the BO strategically proposes new parameter combinations, which are subsequently evaluated by the RS-DNN, to iteratively minimize the discrepancy between the predicted

and target nonlinear mechanical properties. The optimizer settings used for the statistical comparison in Fig. 6 were fixed as follows. GA used an initial population size of 10, 4 generations, top50% elitism, random crossover between elite candidates, and Gaussian mutation with a probability of 0.1, resulting in 40 RS-DNN objective evaluations. BO was performed using a Gaussian-process-based Bayesian optimization framework with 10 initial points, 100 optimization calls, expected improvement as the acquisition function, and a fixed random seed of 42, resulting in 110 RS-DNN objective evaluations including the initial candidates. Together, these three pathways establish an uncertainty-aware inverse design framework that combines generative diversity, surrogate-based physical screening and iterative refinement. This strategy not only mitigates the one-to-many mapping problem but also enables efficient exploration of the disordered design space to identify candidate architectures that better match prescribed nonlinear tensile responses.

4. Results and discussions

This section evaluates the predictive, inverse design, and validation performance of the framework. Sections 4.1 and 4.2 assess forward prediction and inverse design accuracy. Section 4.3 examines statistical structure-property relationships and their mechanical context. Section 4.4 evaluates generated candidates through FEA and periodic references, and Section 4.5 assesses experimental fabricability and qualitative crack path consistency.

4.1. Forward prediction of nonlinear tensile responses via RS-DNN

The ability of the RS-DNN to accurately reproduce the nonlinear tensile responses of the spiderweb-inspired disordered metamaterials is evaluated. The left panel of Fig. 5(a) presents the scatter plot of predicted versus ground truth stress values for all test samples, with NRMSE = 0.0635, revealing a strong linear correlation alongside symmetric and consistent marginal density profiles, thereby illustrating the local predictive accuracy and robust capability of the RS-DNN to capture global statistical distributions. The corresponding right panel illustrates the probability density function (PDF) and the cumulative distribution function (CDF) of the NRMSE for both the training and testing datasets. The NRMSE distributions are narrowly concentrated near zero, with over 90% of the test samples yielding an error below 0.16 and exhibiting closely aligned trajectories across both datasets, indicating robust generalization of the RS-DNN and avoidance of overfitting on unseen metamaterial designs. Furthermore, to evaluate the predictive performance of the RS-DNN, two representative samples from the testing dataset are selected to compare the predicted stress-strain curves with the target curves, as displayed in Fig. 5(b). The results demonstrate NRMSE values of 0.0393 and 0.0866, respectively, thereby confirming the validity of the RS-DNN. To further justify the forward surrogate architecture, the proposed RS-DNN was benchmarked against a standard MLP and a parameter-matched MLP using the same validation-based training protocol. The RS-DNN achieved the lowest full-curve test NRMSE among the three architectures, as shown in Fig. S2. Taken together, these results indicate that the RS-DNN provides a reliable surrogate for nonlinear tensile response prediction across the sampled design space.

4.2. Generative inverse design and refinement

The ability of the inverse-design workflow to recover architectures that reproduce prescribed tensile responses is examined. As depicted on the left panel of Fig. 6(a), the SA-cGAN obtains the optimal inverse design parameters by generating 10 MC dropout predictions per test sample and selecting the most accurate one. The high accuracy of these results is validated by the PDF and CDF of the NRMSE. Meanwhile, the raincloud plot in Fig. 6(a) (right panel) compares the NRMSE

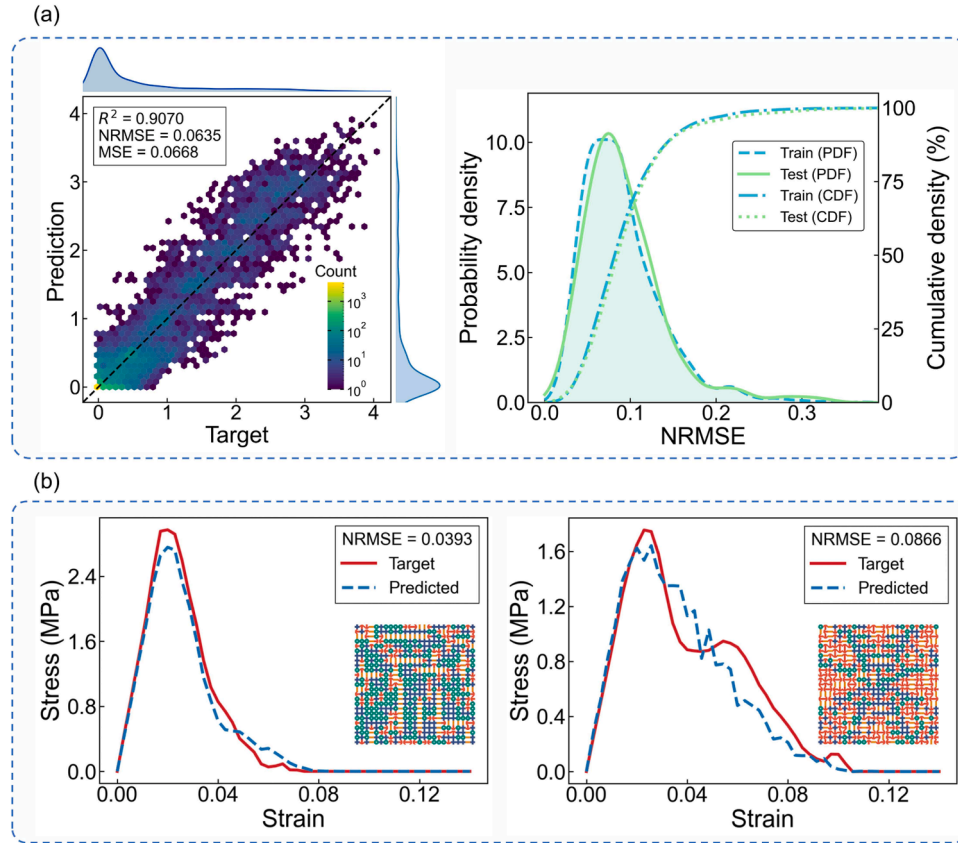


Fig. 5. Evaluation of the RS-DNN. (a) Stress scatter plot, presenting predicted versus target stress values for the test set (left panel). NRMSE distributions, evaluating the probability density function (PDF) and cumulative distribution function (CDF) across datasets to demonstrate model generalization (right panel). (b) Representative stress-strain curves, comparing predicted and target nonlinear mechanical responses to confirm validity of the RS-DNN.

distributions of three inverse design methods (SA-cGAN, GA refinement, and BO refinement), thereby providing a joint assessment of accuracy and robustness. Notably, GA refinement exhibits the best overall performance, achieving the lowest mean and median NRMSE. Its narrow interquartile range and highly concentrated density distribution indicate superior stability across different target configurations. Meanwhile, SA-cGAN serves as a robust baseline with a well-bounded NRMSE distribution. In contrast, while BO refinement can generate high-fidelity solutions for certain cases, it demonstrates the highest overall variance. Specifically, the BO density plot reveals a pronounced upper tail with multiple outliers extending to an NRMSE of 0.24, suggesting greater susceptibility to local optima entrapment or instability when optimizing complex target curves. Taken together, these results indicate that GA refinement offers the most consistent balance of accuracy and stability, whereas BO refinement is more sensitive to the local structure of the design landscape. Because both GA and BO refinements in Fig. 6 are initialized from candidates generated by SA-cGAN, an additional control analysis was performed to determine whether this learned initialization provides an advantage over random initialization. In Supplementary Note S5, random initialization and SA-cGAN initialization are compared under the same RS-DNN surrogate, physical design constraints, and optimizer settings, thereby isolating the effect of the initial candidate source.

To further examine the instability of BO refinement, the optimization trajectory on the right panel of Fig. 6(b) is analyzed. Although the algorithm rapidly reaches a minimum NRMSE of 0.062 immediately following the initial sampling phase, its subsequent iterations exhibit persistent oscillatory behavior, fluctuating between 0.065 and 0.090. This failure to settle into a stable optimal basin suggests a rugged, highly non-convex design space for the spiderweb parameters, and visually

corroborates the high variance of the BO method. The left panel of Fig. 6(b) then presents a representative inverse-design case comparing the three methods. Consistent with the previous statistical analysis, GA refinement achieves the highest accuracy. Notably, despite its overall instability discussed earlier, BO successfully outperforms the baseline SA-cGAN in this specific instance. This result indicates that the three pathways are not simply redundant, but instead offer complementary search behaviors across different target responses. Furthermore, the left panel of Fig. 6(c) visualizes the corresponding optimized microstructures. As observed, the three algorithms generate distinct microstructural layouts for the same target curve. Despite these diverse topological features, all three configurations produce comparable macroscopic nonlinear mechanical responses. This observation directly reflects the inherent one-to-many mapping in metamaterial inverse design [72–74]. The proposed framework does not eliminate this non-uniqueness; it accommodates the issue by identifying multiple feasible designs for a given target response. This capability enriches the design diversity available for subsequent selection under geometric, manufacturing, or application constraints. To quantitatively illustrate this, the radar chart on the right panel of Fig. 6(c) compares the derived parameters alongside specific energy absorption (SEA), and specific strength (SS), both of which are frequently considered metrics for evaluating energy dissipation and load-bearing capacity in lightweight engineering [75]. As depicted, the SEA and SS values for all three strategies align closely, confirming comparable macroscopic performance. However, the underlying geometric and frequency parameters exhibit substantial variations. This parametric divergence explains how distinct microstructural configurations can achieve similar mechanical behaviors, further corroborating the inherent one-to-many mapping issue. Intriguingly, the structure discovered by BO possesses an unconventional topology. This

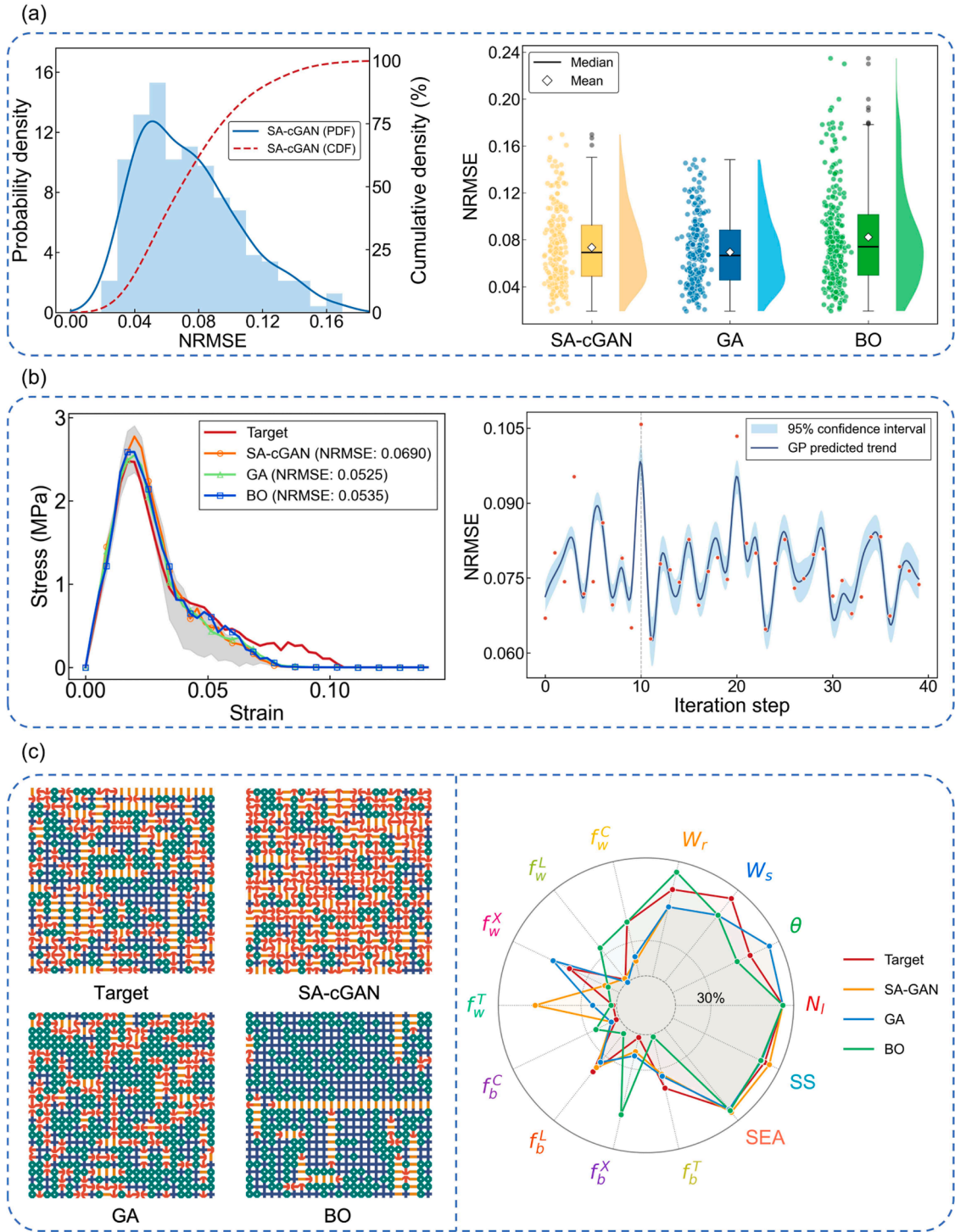


Fig. 6. Comparative evaluation of inverse design pathways and the one-to-many mapping. (a) Statistical performance and robustness, evaluating the NRMSE distributions across the three inverse design methods using all 234 cases from the untouched test set. (b) Representative case and optimization trajectories, comparing the predictive accuracy of the three methods while revealing the fluctuating tendencies of the BO refinement. (c) Microstructural diversity and parametric divergence, visualizing distinct optimized topologies and a radar chart comparison. The panel shows that different geometric and frequency parameter sets can yield comparable macroscopic mechanical responses, and that the framework accommodates the inherent one-to-many mapping and expands design diversity.

reflects the inherent flexibility of the BO algorithm, as driven by the exploration mechanism of its acquisition function, it can actively sample high-uncertainty regions of the design space, allowing it to explore beyond typical design patterns and discover unconventional yet viable structural configurations. In practical terms, such diversity is valuable because it allows secondary design requirements, such as structural connectivity, manufacturability, or energy absorption, to be considered

after the primary tensile target has been satisfied [34,76].

4.3. Interpretable analysis of structure-property relationships for SEA and SS

The inverse design results in Section 4.2 show that different architectures can reproduce similar target responses, but the statistical

relationships between the design variables and SEA or SS remain unclear. In this section, Pearson correlation is used to quantify linear associations, while SHapley Additive exPlanations (SHAP) analysis is used to characterize nonlinear feature contributions within the sampled database. Deformation and fracture morphologies are then examined to provide mechanical context for these statistical trends. The analyses are limited to the prescribed parameterized design space and do not establish material independent causal laws.

Fig. 7(a) shows the correlations between the investigated design parameters and mechanical properties. A positive Pearson correlation (r

$= 0.66$) is observed between SEA and SS within the present database. This coefficient represents a moderate statistical association and does not establish a direct physical coupling or causal mechanism. The subsequent SHAP analysis and fracture morphology are therefore used to examine parameter contributions and provide mechanical context. Among the geometric parameters, the W_s , W_r , and θ show positive correlations with both SEA and SS. In particular, the W_s has the most manifested impact. This occurs because the increased W_s enlarges the area of mapped black pixels, thereby increasing the probability of Cross and X appearing within these regions, as shown in Fig. 8(a). Such

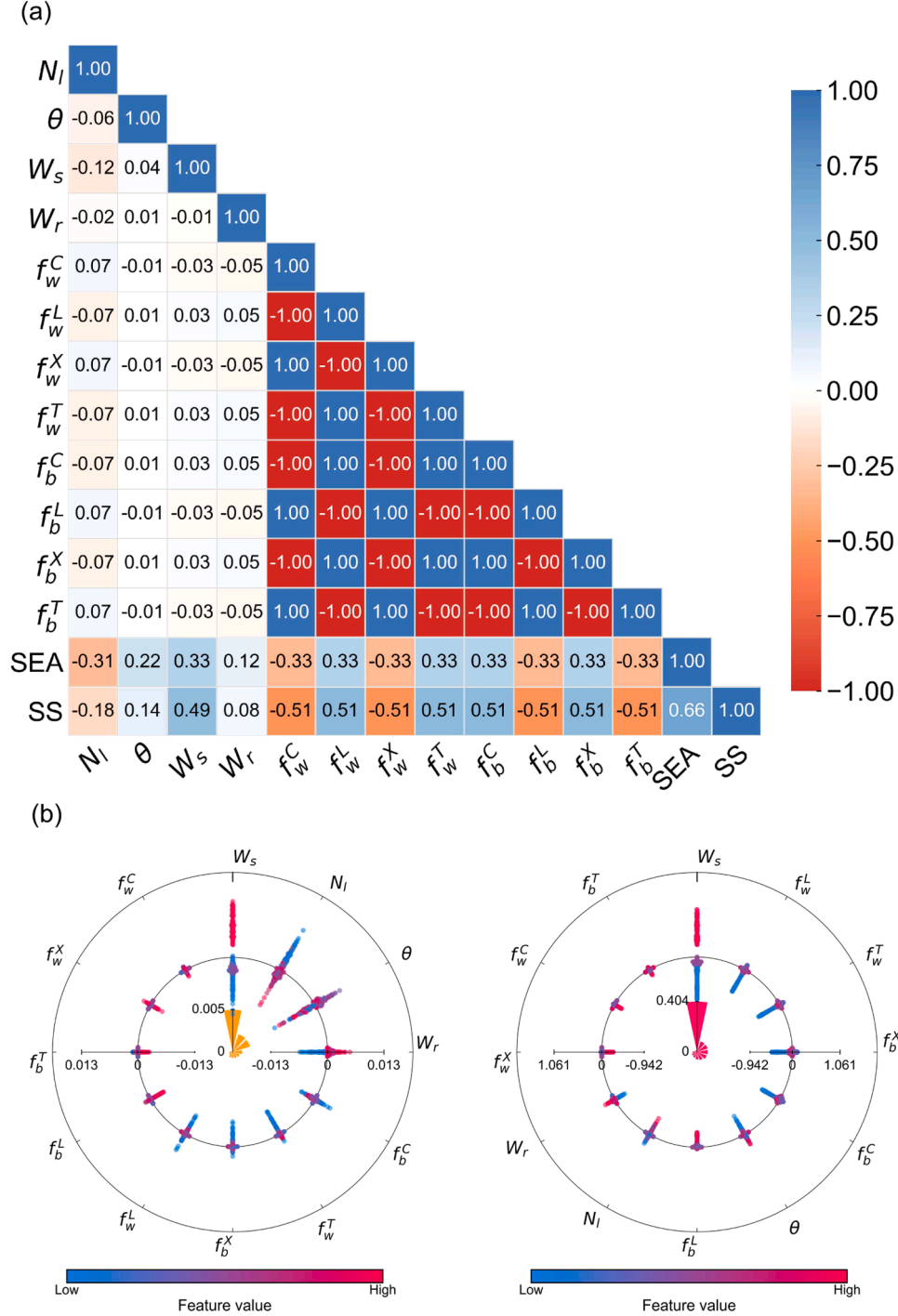


Fig. 7. Interpretable data-driven analysis of structure-property relationships. (a) Pearson correlation, evaluating the linear trends between design parameters and mechanical properties. (b) SHapley Additive exPlanations (SHAP) analysis, quantifying the global importance and nonlinear contributions of individual geometric and frequency parameters to clarify their underlying physical effects on macroscopic performance.

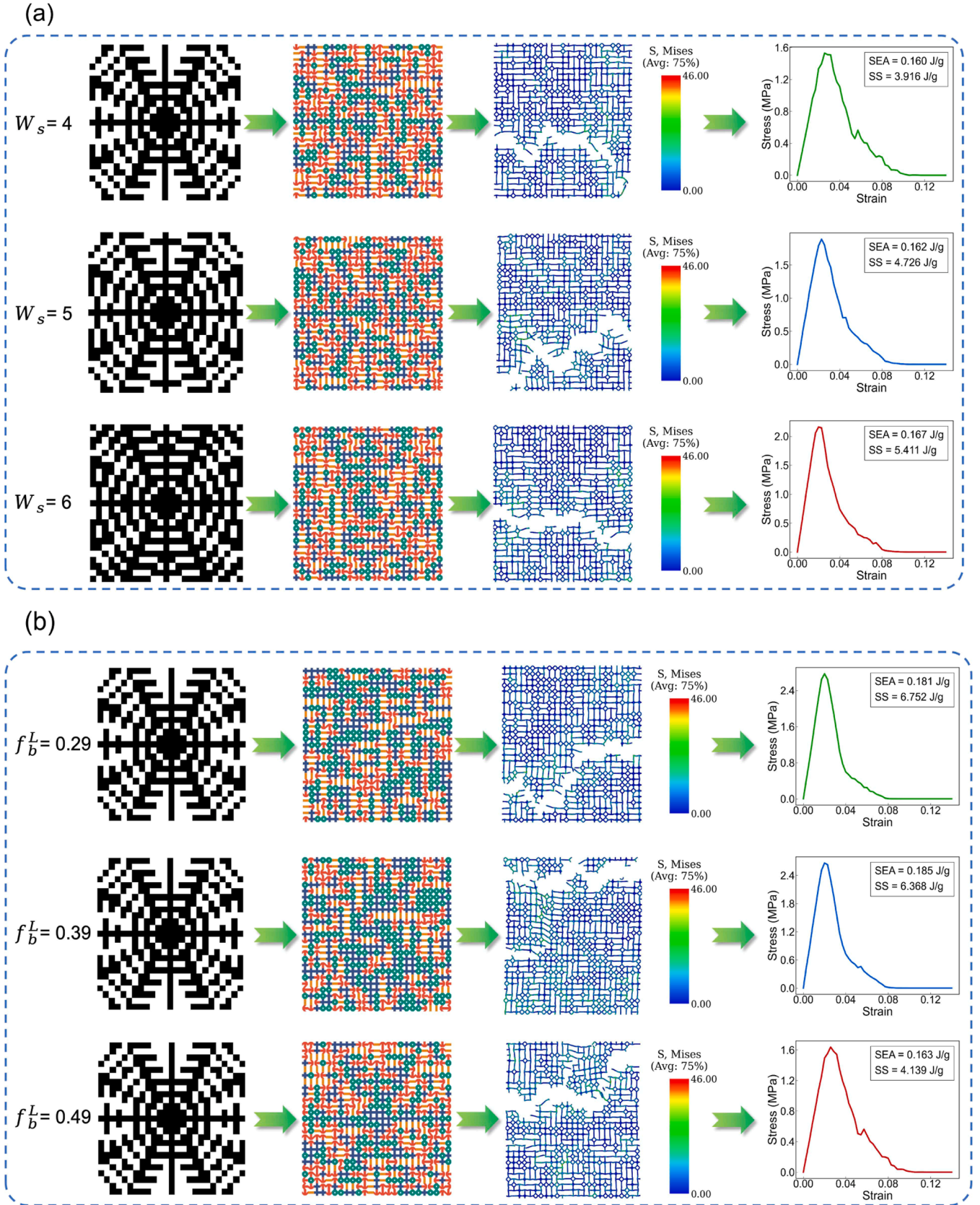


Fig. 8. Physical mechanisms underlying the parametric effects on mechanical performance. (a) Effect of W_s on mechanical performance, illustrating how the increased W_s promotes Cross and X building blocks to form a compact network that facilitates efficient load transfer. (b) Effect of f_b^L on mechanical performance, demonstrating how higher values of f_b^L promote Line and T building blocks to induce uneven voids, leading to local stress concentration and ultimate structural failure.

building blocks, featuring an average coordination number of 4, promote a more compact and uniform network structure that facilitates efficient load redistribution, thereby enhancing both SEA and SS [69, 77]. However, it is also noted that the N_l correlates negatively with both metrics. Aside from the geometric features, the eight frequency parameters display a distinctly alternating correlation pattern ($r = \pm 1$) with each other. This strong linear correlation primarily arises from their mathematical definitions. Specifically, $f_w^L = f_w^T = f_b^C = f_b^X = 0.5 - f_w^C$ and $f_w^X = f_b^L = f_b^T = f_w^C$.

To explain the nonlinear structure-property mappings beyond global linear trends, SHAP analysis was conducted for SEA and SS. Fig. 7(b) presents the SHAP polar honeycomb plots for both metrics, where the global importance of each feature is indicated by the length of its central bar (i.e., the mean absolute SHAP value). Consistent with the Pearson correlations, the W_s ranks first in global importance. It exhibits the widest radial distribution, with high feature values (red dots) largely concentrated in the positive SHAP region, suggesting that the increased W_s contributes to higher predicted SEA and SS. Similarly, the W_r and θ present positive contributions, whereas the N_l exerts a negative effect. Furthermore, SHAP analysis helps clarify the physical effects of the

frequency parameters. For instance, the left panel shows that f_w^T has a stronger positive impact on SEA than f_w^L , whereas the right panel indicates that f_w^L plays a more substantial role in improving SS. In contrast to these differing positive contributions, certain parameters show a consistent suppressive effect. Specifically, higher values of f_b^L negatively modulate performance. This behavior occurs because the increased f_b^L promotes the emergence of Line and T building blocks in the background region, inducing uneven voids that cause local stress concentration and subsequent crack initiation [6,69,77].

The physical effect of this suppressive parameter becomes clearer when connected to fracture morphology. Fig. 8(b) shows that the ultimate structural failure predominantly occurs across the dense clusters of Line and T building blocks. Alongside capturing the fracture trajectory, Fig. 8(b) also reveals a distinct non-monotonic evolution, where the SEA initially increases but subsequently decreases with higher values of f_b^L . To further investigate the underlying reasons for this non-monotonic trend, the respective proportions of the Cross (p_C), Line (p_L), X (p_X) and T (p_T) across the generated structures were analyzed to determine how their relative distributions influence the macroscopic SEA and SS. As illustrated in Fig. 9(a), Pearson correlation analysis was first

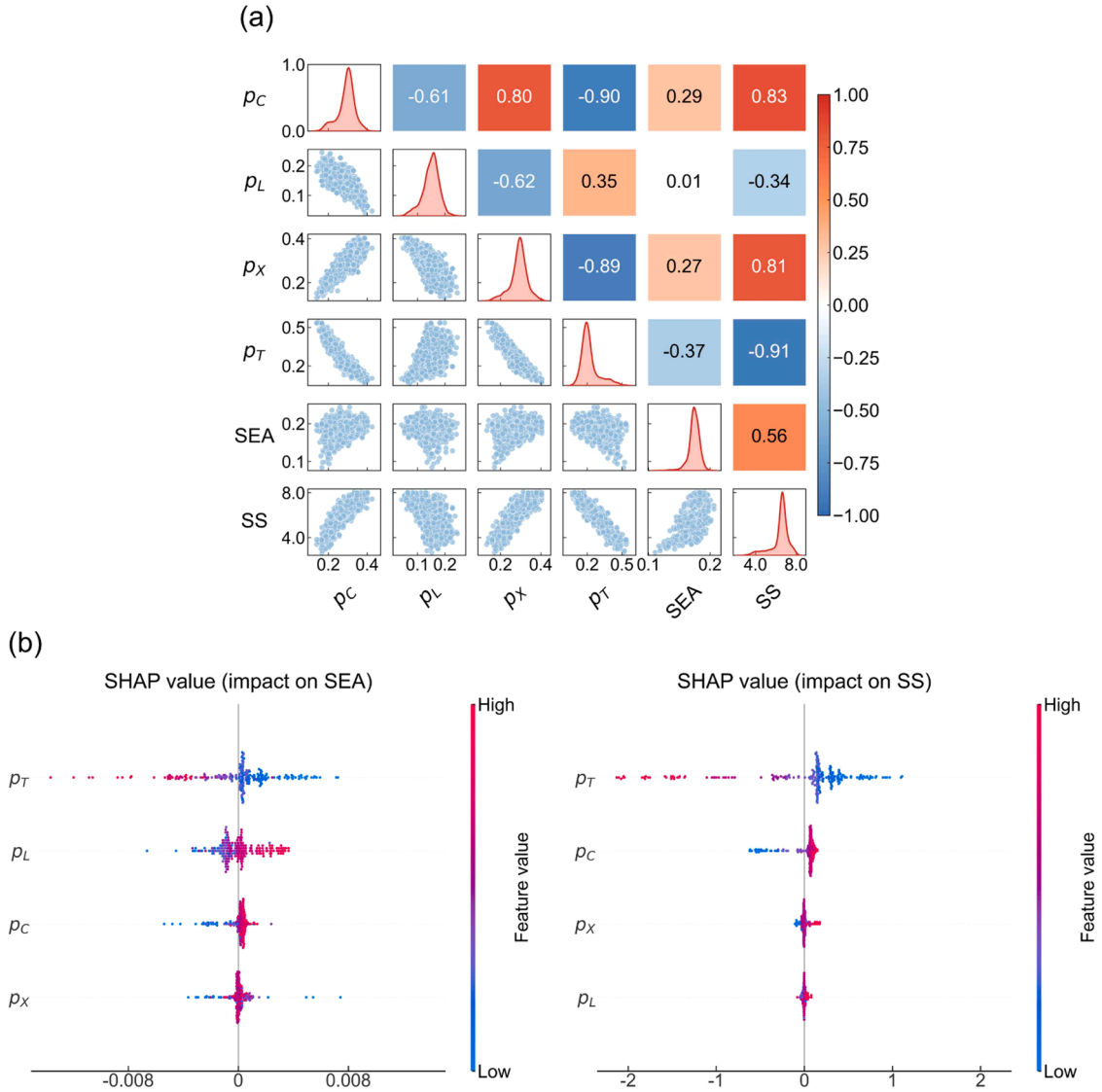


Fig. 9. Impact of building block proportions on macroscopic mechanical properties. (a) Pearson correlation, evaluating the global linear trends to reveal the distinct positive and negative impacts of the four building block proportions on SEA and SS. (b) SHAP analysis, quantifying the individual contributions of these proportions to identify the dominant suppressive and positive factors governing the mechanical performance.

conducted to reveal the global linear trends. Specifically, p_C and p_X exhibit moderate positive correlations with SEA and pronounced positive impacts on SS. In contrast, p_T exerts a moderate negative effect on SEA but a substantial negative effect on SS, whereas p_L shows only a

marginal influence overall. To further quantify the contributions of these four variables, SHAP analysis was employed, as shown in Fig. 9(b). The results indicate that p_T acts as the dominant suppressive factor for both SEA and SS. Meanwhile, p_C and p_X consistently provide positive

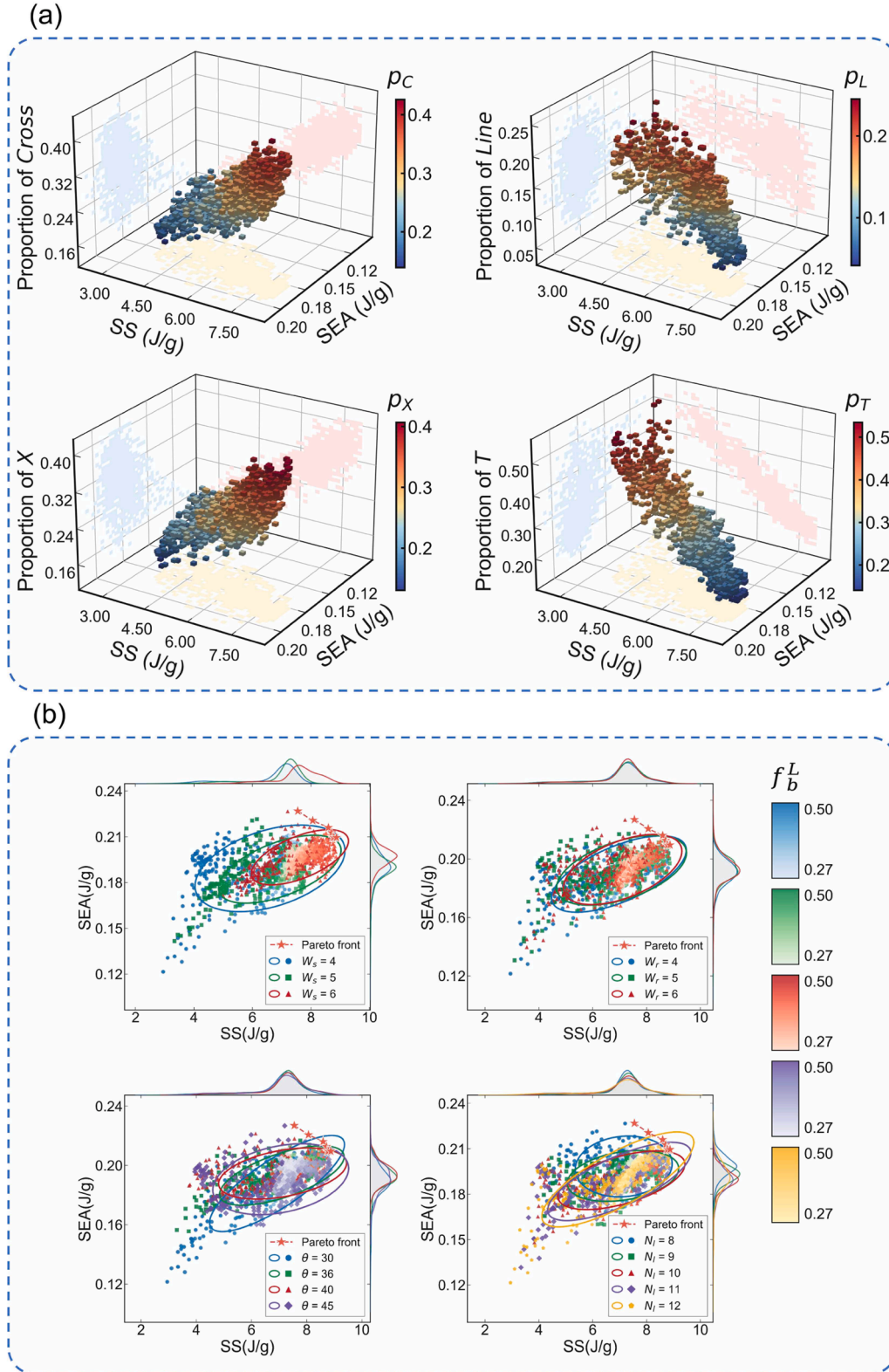


Fig. 10. Parameter-property relationships and mechanical performance distribution. (a) Influence of building block proportions on SEA and SS, the 3D plots show the evolution of SEA and SS against the building block proportions. (b) Multi-parametric performance distribution and parameter-property relationships, visualizing the mapped design space via scatter plots and 95% confidence ellipses.

contributions, and the impact of p_L remains relatively minor. These statistical patterns are also clearly demonstrated in Fig. 10(a), which maps the evolution of SEA and SS against the building block proportions.

These results provide a coherent explanation for the non-monotonic evolution of SEA observed in Fig. 8(b). Specifically, when f_b^L increases from 0.29 to 0.39, p_C and p_X rise from 28.64% and 28.48% to 28.80% and 30.24%, respectively. This positive change helps raise the SEA from 0.181 to 0.185. However, p_T also increases slightly (from 25.28% to 27.52%). Because p_T has a strong negative effect on SS, the SS value drops from 6.752 to 6.368. Furthermore, in the case of $f_b^L = 0.49$, there is a clear change in the component proportions: p_C and p_X drop to 22.24% and 23.04%, while p_T increases to 39.04%, which makes the structure more susceptible to local stress concentrations at the voids

formed by T and Line building blocks. These localized stresses promote subsequent fracture, bringing the SEA and SS down to 0.163 and 4.139, respectively. Therefore, the mechanical response is governed not simply by the increase of any single block type, but by the competitive redistribution among Cross, X, Line, and T units. In general, better performance is associated with increasing p_C and p_X while reducing p_T , because the former enhances structural compactness and load transfer, whereas the latter raises the probability of uneven void formation and severe local stress concentration.

4.4. FEA validation of generated candidates and performance comparison

Following the structure-property analysis, the original design space

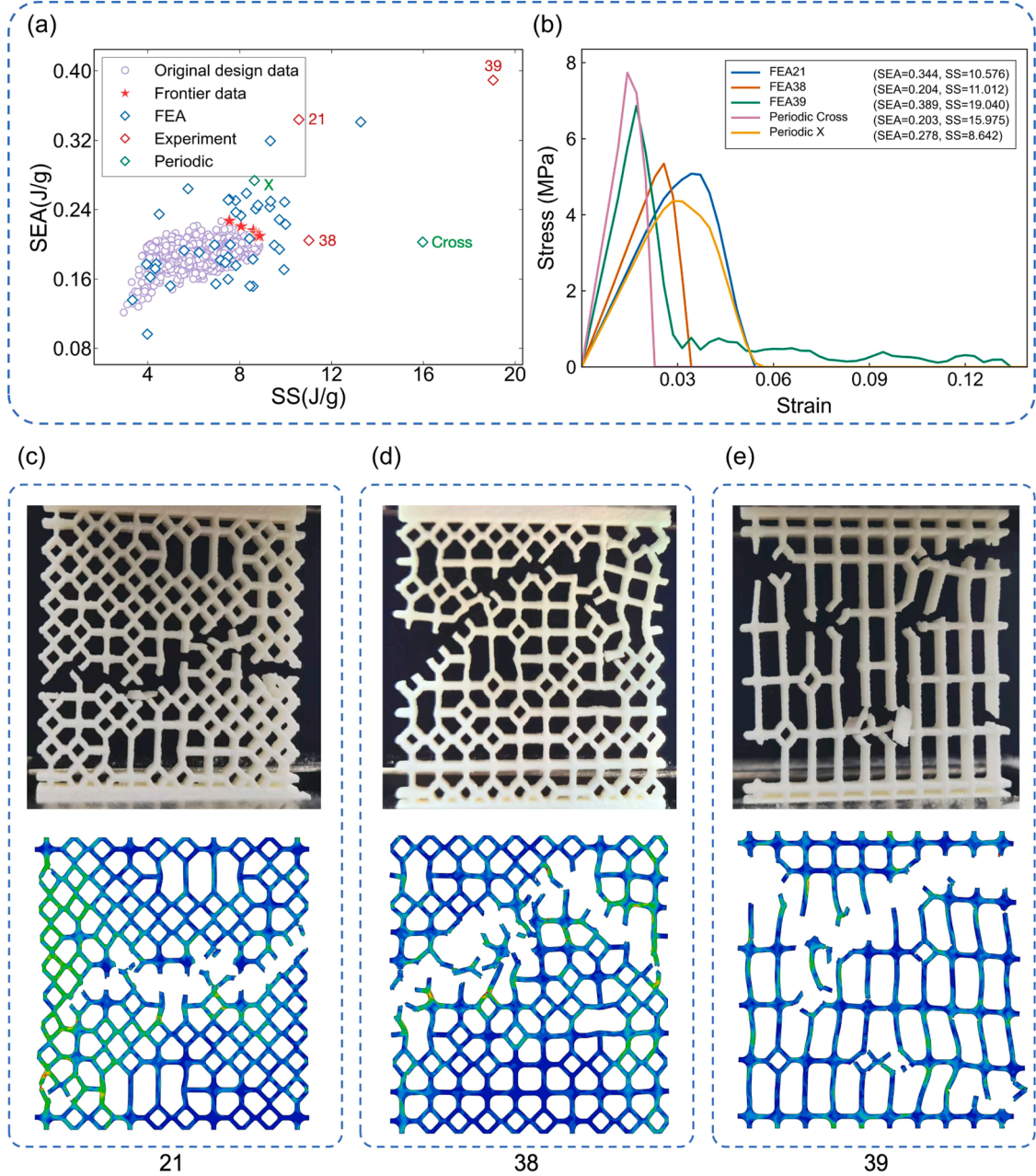


Fig. 11. FEA validation, performance comparison, and experimental crack path assessment of generated candidates. (a) FEA validated SEA-SS distribution, comparing the original dataset, the generated candidates, and periodic Cross and X reference structures. The frontier data are defined by the Pareto front before optimization in Fig. 10(b). (b) Stress-strain curves of representative candidates 21, 38, and 39 and periodic Cross and X structures. (c) Simulated and experimental crack paths of candidate 21. (d) Simulated and experimental crack paths of candidate 38. (e) Simulated and experimental crack paths of candidate 39. Panels (c-e) are used to compare the main failure locations and crack propagation tendencies in scaled printed specimens.

was further examined to establish a reference for candidate generation and FEA validation. Fig. 10(b) maps the SEA-SS distribution of the original database and groups the architectures according to W_s , W_r , θ , and N_f . The corresponding 95% confidence ellipses summarize the performance ranges associated with different parameter levels, while the color intensity within each group represents f_b^L . The upper-right boundary of the distribution defines the pre-optimization Pareto front, which is used as the frontier data in the subsequent search [78–80]. The grouped distributions are also consistent with the global trends identified in Section 4.3. For example, as W_s increases from 4 to 6, the corresponding confidence ellipse shifts toward higher SEA and SS. Across the four panels, points with larger f_b^L values are concentrated mainly in the lower-left region, whereas smaller f_b^L values occur more frequently near the upper-right boundary. This distribution supports the negative association of f_b^L with SEA and SS within the prescribed database. These relationships describe global trends under the original constrained parameterization and are not used as deterministic rules for individual generated candidates outside the sampled frequency relationships.

Based on the frontier data extracted from the pre-optimization Pareto front in Fig. 10(b), candidate generation was performed through sequential generation and refinement. The SA-cGAN integrated with MC dropout was first used to generate candidate parameters, and GA and BO were then used to refine the search with RS-DNN as a fast performance evaluator. During this process, a candidate was retained only when the RS-DNN-predicted SS and SEA were both improved by at least 0.5% and no more than 4.5% relative to an existing frontier data point. This criterion was used as a bounded candidate screening rule. The final mechanical performance of each generated candidate was determined by subsequent FEA validation.

Fig. 11(a) presents the FEA validated SEA-SS distribution, including the original dataset, the 41 generated candidates validated by FEA, and two periodic reference structures constructed from Cross and X building blocks. Among the 41 generated candidates, 20 remain outside the frontier data after FEA validation. The validated distribution suggests an asymmetry in search difficulty. Candidates with balanced improvements in both SEA and SS are more difficult to identify than candidates with a pronounced improvement in one metric and a smaller improvement in the other. Candidate 39, which shows the highest FEA validated SEA and SS among the 41 generated candidates, was captured by BO. This result shows that BO can access a sparsely sampled region with high values of both metrics under the present search setting. However, the statistical comparison in Section 4.2 shows that GA provides more stable refinement across the evaluated target cases.

The periodic reference structures provide a direct baseline for evaluating the practical significance of the extended SEA-SS range. The periodic Cross structure exhibits a relatively high SS but a low SEA, whereas the periodic X structure exhibits a higher SEA but a lower SS. Several FEA validated disordered structures achieve more favorable SEA and SS combinations than these periodic references within the present spiderweb parameterized design framework. In particular, candidate 39 improves SEA by 91.63% and SS by 19.19% relative to the periodic Cross structure. Relative to the periodic X structure, SEA and SS are increased by 39.93% and 120.32%, respectively.

Fig. 11(b) compares the stress-strain curves of candidates 21, 38, and 39 with those of the periodic Cross and X structures. These three candidates were selected to represent different FEA validated SEA and SS combinations. Candidate 21 exhibits a relatively high SEA while maintaining a moderate SS. Candidate 38 has a higher SS than candidate 21 but a lower SEA, whereas candidate 39 combines the highest values of both metrics among the 41 candidates. The corresponding geometries of these candidates and the two periodic reference structures are provided in Supplementary Note S7.1.

Although 20 FEA validated candidates remain outside the original frontier data, the performance predicted by RS-DNN does not always agree closely with the FEA results. A broader error analysis was

therefore conducted for all 41 generated candidates. The curve NRMSE and the relative errors of SS and SEA for each candidate are reported in Table S1 of Supplementary Note S6. This analysis defines the current reliability of RS-DNN when generated candidates approach or exceed the performance boundary of the training database.

This limitation motivates the integration of active learning with uncertainty based screening. Candidate structures predicted beyond the current performance boundary can first be screened using uncertainty or distance criteria and then validated by FEA. Candidates with FEA validated high performance, together with candidates showing large surrogate errors, can subsequently be added to the RS-DNN training set. Repeated cycles of prediction, screening, FEA validation, and model updating would progressively expand the reliable prediction boundary with a limited number of additional FEA samples [81]. In addition to updating the surrogate through active learning, the prescribed frequency relationships could be progressively relaxed to expand the design space. Independent variation of the regional block frequencies would increase the available block compositions and introduce load paths and deformation patterns that are excluded from the current database. The attainable SEA-SS range may consequently be shifted or broadened, with different architectures favoring load bearing capacity, energy absorption, or a balance between the two properties. Additional FEA data and corresponding updates to the surrogate and inverse design models would be required to determine the resulting performance boundaries.

Finally, the reported SEA-SS trends are specific to the adopted PA12 nylon material model. The identified structural factors, including connectivity, load path continuity, regional block composition, and local stress concentration, may remain qualitatively relevant to other material classes. However, the quantitative performance range, optimal frequency distribution, and dominant failure mode depend on the constitutive and fracture behavior of the base material. Extension to ductile metals or elastomers will require material specific calibration of the FE model, regeneration of the mechanical database, and retraining of the surrogate and inverse design models.

4.5. Experimental feasibility and qualitative crack path assessment of representative scaled architectures

To evaluate the practical feasibility of the generated architectures, candidates 21, 38, and 39 were selected for fabrication and tensile testing. PA12 nylon specimens were fabricated using Multi Jet Fusion (MJF) on an HP Jet Fusion 5600 system. Because the thin walls and small pores of the original 25×25 architectures were below the reliable printing resolution of the available MJF process, 10×10 architectures were adopted and enlarged by a factor of 2.5. Each specimen had overall dimensions of $30 \text{ mm} \times 30 \text{ mm} \times 5 \text{ mm}$. For each selected architecture, three nominally identical specimens were fabricated and tested. Tensile tests were conducted using an LE3000 universal testing machine with a maximum load capacity of 50 kN at a loading rate of 1 mm/min.

The reduction in spatial resolution alters the spiderweb mapping and the local distribution of building blocks. Therefore, the scaled 10×10 specimens do not fully reproduce the original 25×25 design space. The experiments are used mainly to assess printing feasibility and qualitative crack path consistency. As shown in Fig. 11(c–e), the main crack locations and propagation directions observed experimentally are generally consistent with the corresponding FEA results.

Among the tested candidates, candidate 39 exhibits the highest FEA validated SEA and SS among the 41 generated candidates. The structural morphology and FEA response were therefore examined to provide a possible explanation, while the experimental observations were used only to assess the predicted crack path. As shown in Fig. 11(e), many connected members are oriented approximately along the loading direction, forming continuous paths that support load transfer. The FEA stress-strain curve in Fig. 11(b) also shows a prolonged response after the peak. This response is consistent with progressive failure and

rotation of multiple members, allowing deformation to continue over a wider strain range and increasing energy absorption. The volume fraction of candidate 39 is 27.90%, lower than those of candidates 21 (34.98%) and 38 (34.63%). For structures with the same overall dimensions and base material, the lower volume fraction corresponds to a lower structural mass. Because SS and SEA are normalized by mass, this reduction can increase SS if the peak load bearing capacity is retained and can increase SEA if the stress level and deformation capacity, including the response after the peak, are maintained. However, the mechanical response cannot be explained by volume fraction or overall block proportions alone. The orientation, spatial arrangement, and connectivity of the building blocks also affect the formation and continuity of load paths and the resulting deformation process. Therefore, the high specific performance of candidate 39 may result from the combined effects of continuous load paths that maintain load bearing capacity, progressive deformation after the peak that contributes to energy absorption, and the reduced mass associated with its lower volume fraction. Controlled structural comparisons and local energy distribution analyses are still required to quantify the contribution of each factor.

Overall, the experiments establish the fabricability of the scaled architectures and provide qualitative support for the simulated crack paths. Further quantitative validation will require calibrated material and fracture parameters, 3D models matching the specimen geometry, and explicit representation of printing induced geometric deviations [82,83]. Further details of specimen scaling, material characterization, stress-strain comparison, and modeling limitations are provided in Supplementary Note S7.

5. Conclusions

A spiderweb inspired framework has been presented for the inverse design of disordered metamaterials with programmable nonlinear tensile responses. By combining the VGA with an ensemble deep learning workflow, this study establishes a data driven route to explore tensile mechanics in disordered architectures and to investigate the balance between SEA and SS within a prescribed spiderweb parameterized design space. The main conclusions are as follows.

First, the RS-DNN provides an efficient forward surrogate for predicting nonlinear tensile stress-strain responses and associated mechanical metrics of spiderweb-inspired metamaterials. This surrogate markedly reduces the computational cost relative to high-fidelity finite element simulations, making large-scale screening and optimization of disordered architectures feasible.

Second, interpretable machine-learning analyses, including Pearson correlation and SHAP analysis, identify how geometric and frequency parameters govern SEA and SS. These results not only quantify the relative importance of key design variables, but also connect data-driven trends to underlying fracture mechanisms, thereby providing physically interpretable guidelines for the design of disordered metamaterials.

Third, the SA-cGAN, combined with MC dropout and subsequent refinement using GA and BO, enables inverse design under the inherently one-to-many mapping between structure and response. This strategy generates diverse and physically viable candidate microstructures for prescribed nonlinear tensile responses. After FEA validation, 20 of the 41 generated candidates remain outside the pre-optimization Pareto front. Within the prescribed spiderweb parameterized framework, the validated nonperiodic candidates exhibit more favorable SEA-SS combinations than the periodic Cross and X architectures. The practical feasibility of the generated designs is further supported by tensile tests of three representative scaled architectures, which confirm their fabricability and show qualitative agreement between the simulated and experimental crack paths.

Overall, the proposed framework provides a systematic and computationally efficient approach for the design of lightweight disordered metamaterials with enhanced strength-toughness synergy within the prescribed spiderweb parameterized design space. More broadly, it

shows how bioinspired architecture generation, interpretable learning and uncertainty-aware inverse design can be integrated to move beyond conventional performance boundaries. Future work will therefore focus on improving surrogate robustness in extrapolative regimes by using active learning to incorporate newly validated candidates with high performance, while progressively relaxing the frequency constraints to expand the reliable design space of disordered metamaterials. In parallel with this design-space extension, applications to metals or elastomers will require material-specific FEA calibration, dataset regeneration, and retraining of the surrogate and inverse design models.

CRediT authorship contribution statement

Haotian Jin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dazhi Zhao:** Writing – review & editing, Visualization, Software, Methodology. **Yujie Xiang:** Writing – review & editing, Visualization, Methodology. **Peng Zhang:** Writing – review & editing, Methodology. **Xianyan Chen:** Writing – review & editing, Software. **Keke Tang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Xianqiao Wang:** Writing – review & editing, Visualization, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work was funded by the National Natural Science Foundation of China (No. 11872278), the Natural Science Foundation of Shanghai (Grant No. 25ZR1401355), and the Fundamental Research Funds for the Central Universities.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijmecsci.2026.111961](https://doi.org/10.1016/j.ijmecsci.2026.111961).

Data availability

Data will be made available on request.

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